

Stage 1 Tutorial Activity

Why Software Engineering Still Matters

Activity 1 -- Think-Pair-Share

If ChatGPT or Copilot can produce a 100-line Python application very quickly, what knowledge does a software engineer still need?

1. Code comprehension
2. Code security
3. Practical application to code [knowing what implementation to use for the given problem]

Activity 2 -- Is this Software Engineering?

Scenario A: A student writes a 50-line Python Calculator

Scenario B: A team develops a payroll system used by 5,000 employees

Scenario C: An AI assistant generates a simple appointment application from one prompt

Scenario	Programming?	Software Engineering?	Why
A	Yes	Not really	Software Engineering refers to applying systematic principles to design and build. Just programming doesn't mean that there was any planning there
B	No	Yes	A team was utilized to create the payroll system. With the team, there is planning, code reviewing and

			things that apply in the Software Engineering lifestyle
C	No	No	AI Assistant doesn't plan for the project or manage a team. It's just given prompts and spits out snippets of code from the internet creating some sort of program. It doesn't really program either, just copy and pastes code snippets from the internet.

Activity 3 -- SmartCare Problem Analysis

Client statement: SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The clinic wants new software to improve these processes.

Task 1 -- Identify stakeholders

Stakeholder	What do they need?
Patients	An efficient program that can book and reschedule appointments quickly and safely.
Receptionists	Fast way to check patients in and avoid double-bookings. Assistance in rescheduling appointments.
Doctors	Reliable access to patients' medical history and schedule.
Owner	Reports on the number of patients.

Task 2 – Identify current problems

1. No real-time access [unless it's in google spreadsheets or something]
2. Poor security [easy to access and easy to copy]
3. No version control [can't revert changes]
4. Potential scheduling conflicts [hard to see how many patients booked in a day]

Task 3 -- Ask client questions

1. How many patients on average per day [or week/month. Just give a timeframe]
2. Who needs to access the systems
3. What operating systems is going to be used
4. What's the budget and timeframe

Activity 4 -- Critique an AI Response

An AI assistant suggests appointment management; facial-recognition login; AI diagnosis recommendations; patient search; online payment; practitioner schedule view; insurance processing; automatic treatment-plan generation.

Suggestion	Client evidence?	In scope?	Decision
Appointment management	Yes [need it for the software]	Yes	Keep
Facial recognition login	No [why would you need this?]	No	Remove
AI diagnosis recommendations	No [terrible idea]	No	Remove
Patient search	Yes [in every database basically I think]	Yes	Keep
Online payment	Yes [I feel like everything is online payment these days]	Yes	Keep
Practitioner schedule view	Yes [of course]	Yes	Keep
Insurance processing	No [has nothing to do with me]	No	Remove
Treatment-plan generation	No [don't know what this is but it's not supposed to be here]	No	Remove

Exit question

Write one activity that a software engineer must perform and that cannot safely be delegated entirely to AI.

Code security is very important for software engineers. The moment this program gets compromised, the severity of the damage to the customers and business reputation will be incomprehensible. An AI cannot manage code security by itself, the code snippets that the AI may utilise can potentially have major issues in it and it is the software engineer's responsibility to clean up the mistakes.